Shiny rrisk user guide

Your quick start to perform probabilistic risk modelling

Getting started

> **Note**
>
> **This section will introduce you to:**
>
> - Description of the main interface
> - Loading an "shiny rrisk" file that contains the the complete documentation and executable code of your model.

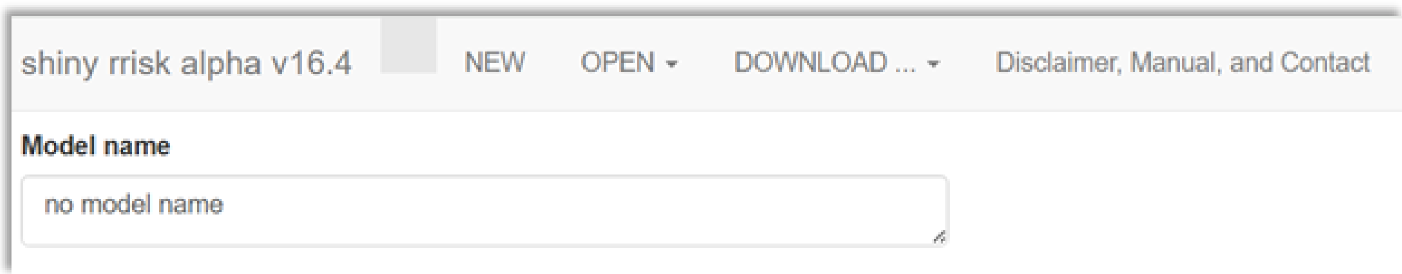


Figure 1: Main screen, top row of shiny rrisk.

The top row of the app displays the version of the app and provides basic functionality for managing models (Figure 1).

NEW	Check here if you wish to create a new model and assign a Model name.
OPEN	Open a saved model or one of the demo models implemented in the tool. The extension of model files is `"rrisk"`.
SAVE	[@Robert[: consider change of button name]{.mark}] Save the model, results or the model report to your local system.
ABOUT	[@Robert[: consider change of button name]{.mark}] Check out here for important information about the app .

> **Avoid loss of your data**
>
> The app runs on a server provided by the German Federal Institute for Risk Assessment (BfR). Your data is not accessible to BfR or any other party and irretrievably lost upon termination of the browser session or interrupt of the server connection. It is highly recommended that you **Save** your model regularly after each important modification.

> **Good practice**

> **Read more**

Model creation

> At the end of this section, you will be able to create a model in shiny rrisk.

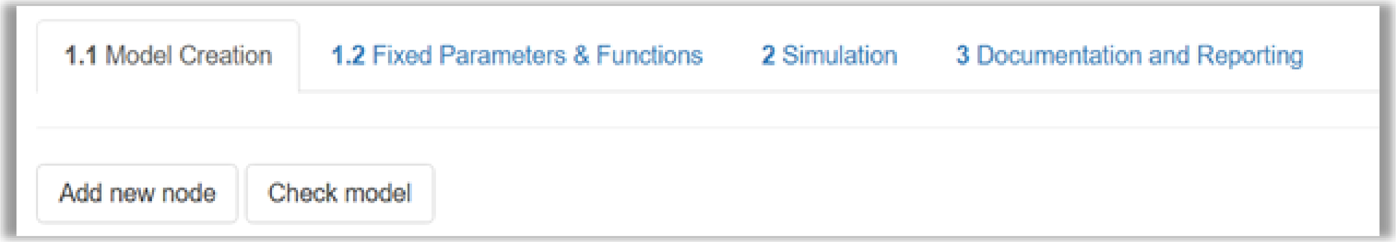


Figure 3: Main screen, second row of shiny rrisk.

A model in shiny rrisk consists of so-called nodes. The user may also define fixed parameters and functions available for the definition of nodes (see @FixedParametersFunctions).

Add node	[@Robert[: consider change of button name]{.mark}] Create a new node and launch its verbal description and mathematical definition.
Check model	Verify that a simulation model can be built from the mathematical and numerical model input you have provided.

Avoid loss of your data
Remember to save your model after adding each node.

Good practice

Read more

Verbal definition of a node

[@MG[: consider change to Add Node: Node Definition - Not sure what it means Verbal]{.mark}]

Read here how easily you will generate an integrated documentation of each node.

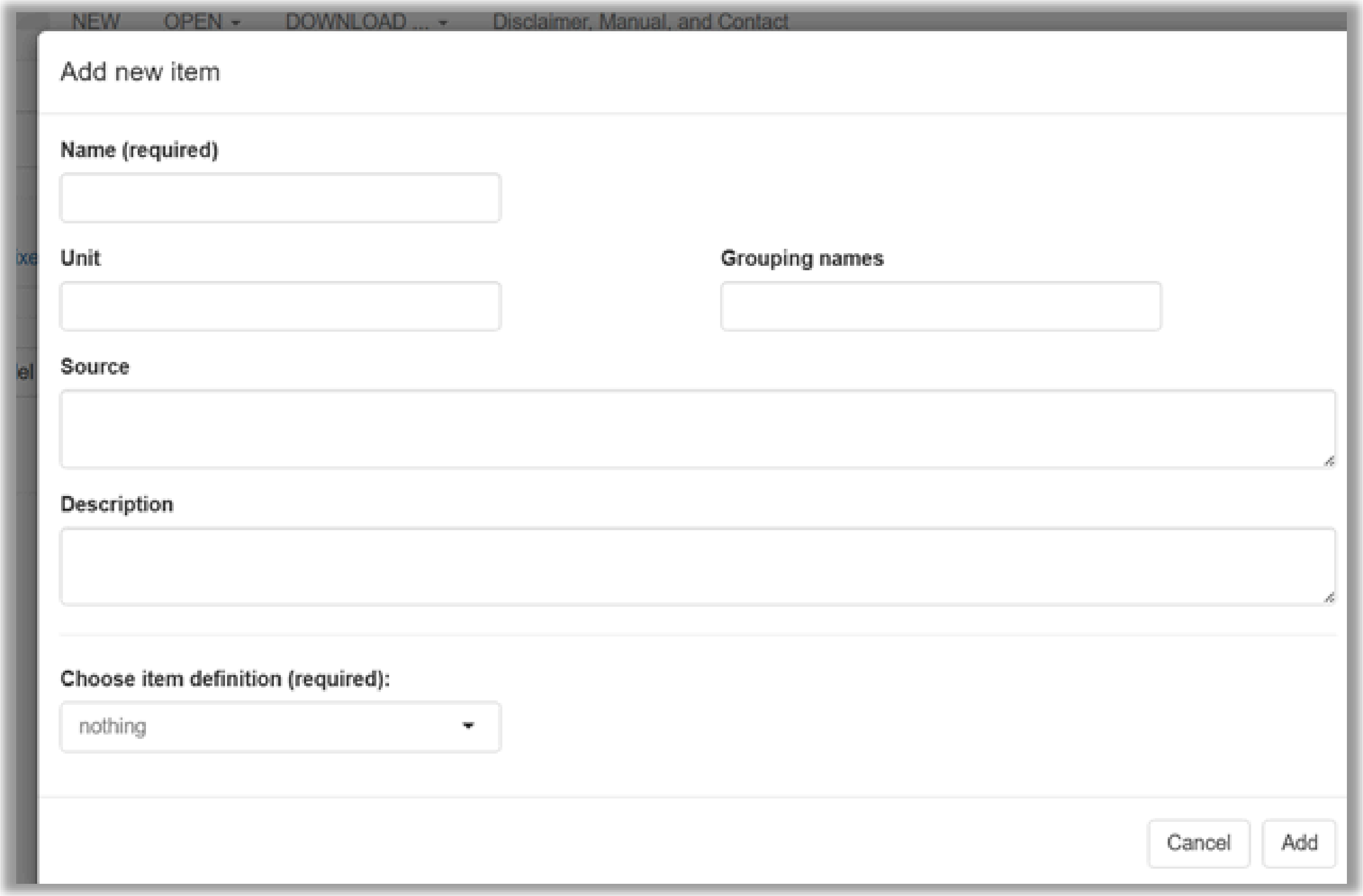


Figure 4: Descriptive part of node definition.

[@MG[: The figure should have "Add Node" instead of Add new item]{.mark}]

Upon launching Add node, a window pops-up with several free-text fields.

Name* @MG here you can mention editing the name of a node	[@Robert[: consider change of button]{.mark}] Each node has to have a unique name starting with a letter. Numbers and underscore are allowed. An error message occurs upon missing or invalid entry.
Unit	The physical measurement unit should be entered here if applicable.
Grouping	[@Robert[: consider change of button name]{.mark}] A grouping name may refer to a particular module of the model or step in the event tree. It is possible to define membership of one node to more than one group in which case the group names are comma-separated. The groups are visualised in the graphical representation of the model and are used to organise the model documentation.
Source	The source of information about the node should be provided according to good scientific practice. If a bibliographic reference is available it should be entered using the format "Author(s), Year (doi code)".
Description	A verbal account of all relevant information required for documentation and review should be provided. This may include any underlying assumptions and uncertainties. The text may be structured using sub-headers (see example for markdown syntax below).
*	Required

Avoid loss of your data
Remember to save your model after adding nodes and/or making important changes in the model

Example for markdown syntax (Alizarin red S in Eel)

Good practice

Read more

Mathematical definition of a node

You will understand how to define a node in shiny rrisk, which is the essential tasks in model creation.

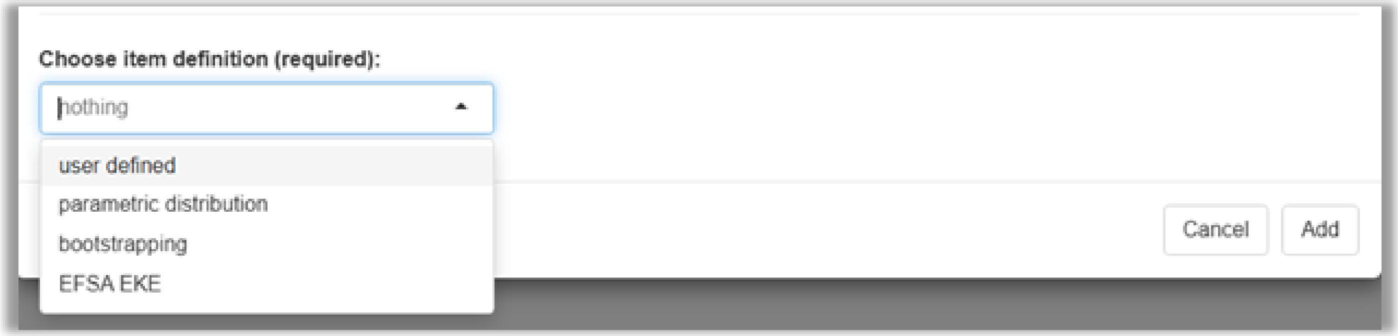


Figure 6: Mathematical part of node definition.[@Robert[: consider change to "Choose node defintion*"]{.mark}]

Several options are available for defining the mathematical properties of a node, including user defined, parametric distribution, bootstrapping and EFSA EKE.

`@MG [: maybe change EFSA EKE to just expert elicitation]`

The first option allows you to define a node using a mathematical expression (Figure 7).

User defined

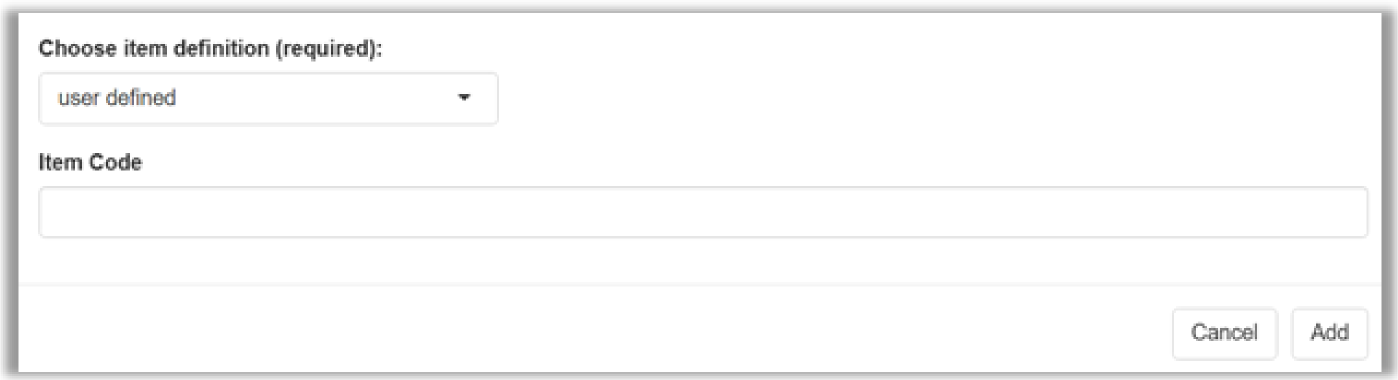


Figure 7: Choose user defined to enter a mathematical expression.

Expression	[@Robert[: consider change button name]{.mark}]A mathematical expression is entered here that can be evaluated similar to a line of code in the software R. At least one term of the expression is another (distribution) node, which is referred to as a “parent node”. Parent nodes that do not yet exist in the model are referred to as “implicit nodes” and will be generated automatically by shiny rrisk.

Example for an expression using a parent node (Alizarin Red S in eel model)

Example for an expression using implicit nodes

Good practice

Read more

Parametric distributions

@MG [re-consider this name since some of the distributions are empirical and therefore non-parametric (cumulative, histogram, general)]

Upon completing this section, you will master the use of probability distributions in in shiny rrisk, a prerequisite for reflecting variability and uncertainty in your model.

@MG [**variability and uncertainty can be reflected in general under the mathematical definition of a node, since it won't only apply to the distributions, but also to formulas using nodes represented by a distribution**]

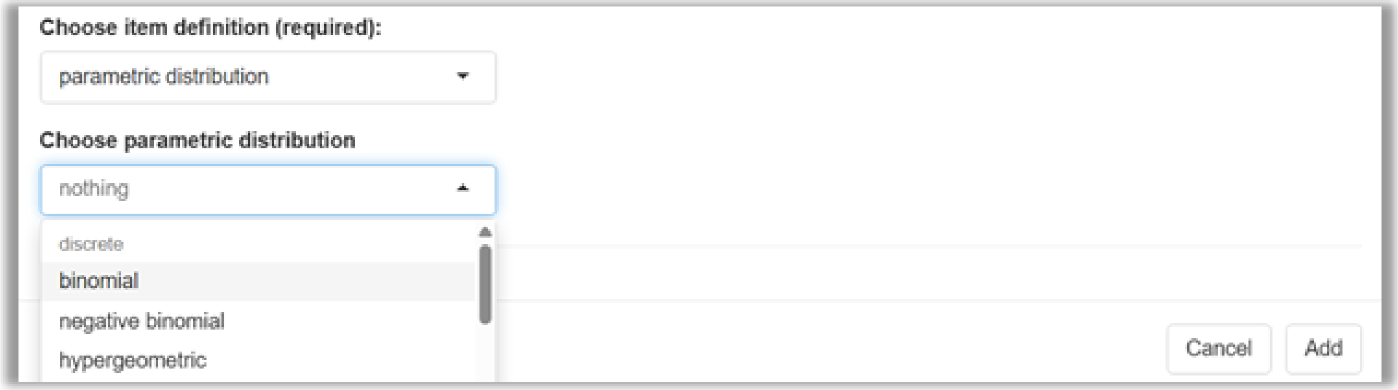


Figure 9: Choose a distribution family.

Once you have selected parametric distributions you can choose from a number of discrete and continuous distribution families (Figure 9).

Choose parametric distribution	Select here the distribution family for the new node.

Discrete distributions

The following five discrete distributions are available to generate a probabilistic node in shiny rrisk: discrete distributions: binomial, hypergeometric, Poisson, negative binomial and discrete.

Continuous distributions

The following thirteen continuous distributions are available to generate a probabilistic node in shiny rrisk: uniform, beta, modified PERT, exponential, Gaussian, lognormal, Weibull, gamma, inverse gamma, shifted log-logistic, triangular, general and cumulative.

Special distributions

Shiny rrisk provides two special distributions. The sigma distribution is …. Please complete. The Furry process (Only birth) is a stochastic process where the probability of an event occurring at a given time depends on the number of events that have already occurred and the rate of new events is proportional to the current population size. The documentation of these distribution is found in … Javier's model if we include it in the demo package..

Variability and uncertainty

The user defines whether a distribution characterises variability or uncertainty. Variability is a reflection of a feature that varies among individuals in a population, e.g., body weight of exposed children of a defined age group (see HEV in liver sausages example below). This should be distinguished from uncertainty, which is a quantification of the limited knowledge about a population statistic, which is an unknown single value, e.g. the true mean body weight of individuals in the defined age group and region. Read more about this topic below.

Context-sensitive distribution parameters

Upon launching any of the distributions, a context-sensitive interface appears for completing the definition of required model parameters. This is illustrated using the the modified PERT distribution in the HEV in liver sausages example below.

Distribution fitting

A shiny rrisk module for fitting distributions is currently being developed.

Known parameter values versus parameter uncertainty

Distribution parameters can be entered as single numbers (see HEV in liver sausages example below). A single number (scalar value) implies that the true parameter value is exactly known. In this case, the distribution reflects variability under the simplifying assumption that both the distribution family as well as the distribution parameters are exactly known. In many cases, however, an uncertainty distribution is required to reflect uncertainty about the parameter value(s) or even a functional dependency of the parameter on other nodes. The latter is illustrated with the L.monocytogenes in cheese example model.

Lower and/or upper limit for truncated distribution

In shiny rrisk, many distributions can be defined in a truncated version by defining a lower and/or upper limit.

Example for parametric distribution (HEV in liver sausages)

Example for a distribution parameter with functional dependency (L.monocytogenes in cheese)

Read more

Bootstrapping

In this section you can explore how to use bootstrapping for generating distributions that represent statistical parameter uncertainty without specifying a distribution family.

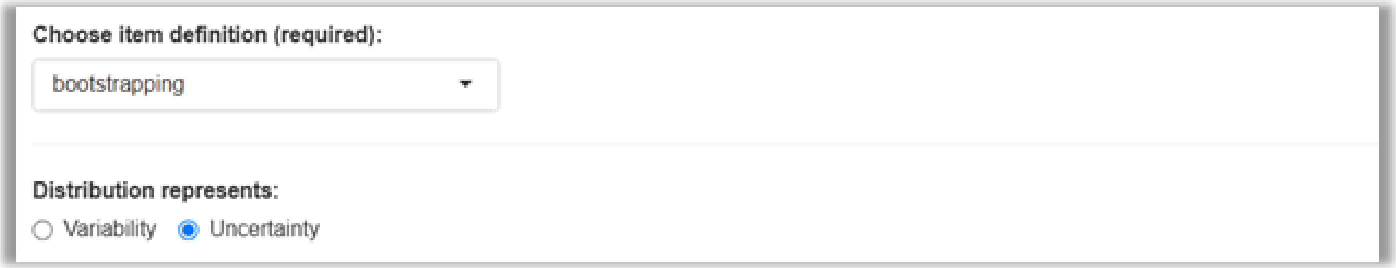


Figure 13: Bootstrap nodes always represent parameter uncertainty.

Non-parametric bootstrapping is available to generate uncertainty distributions for one or more statistics of a given data set (Figure 13). Details are illustrated using the example below.

Application of bootstrapping

The application of bootstrapping in shiny rrrisk involves three steps.

1. Defining a bootstrap node, which involves documenting the data source and defining node names and corresponding statistics.
2. Generating a bootstrap distribution node for each statistic requested in step 1 (automatically by shiny rrisk).
3. Using the distribution node(s) generated in step 2 as parent nodes in the definition of further nodes.

Example for non-parametric bootstrap (Alizarin red S in Eel model)

Read more

Expert knowledge elicitation (EKE)

This section demonstrates how you can use typical results from an expert elicitation to parameterise a node in shiny rrisk Needs major revision when the distribution app becomes available; Olaf Mosbach-Schulz should review the section.

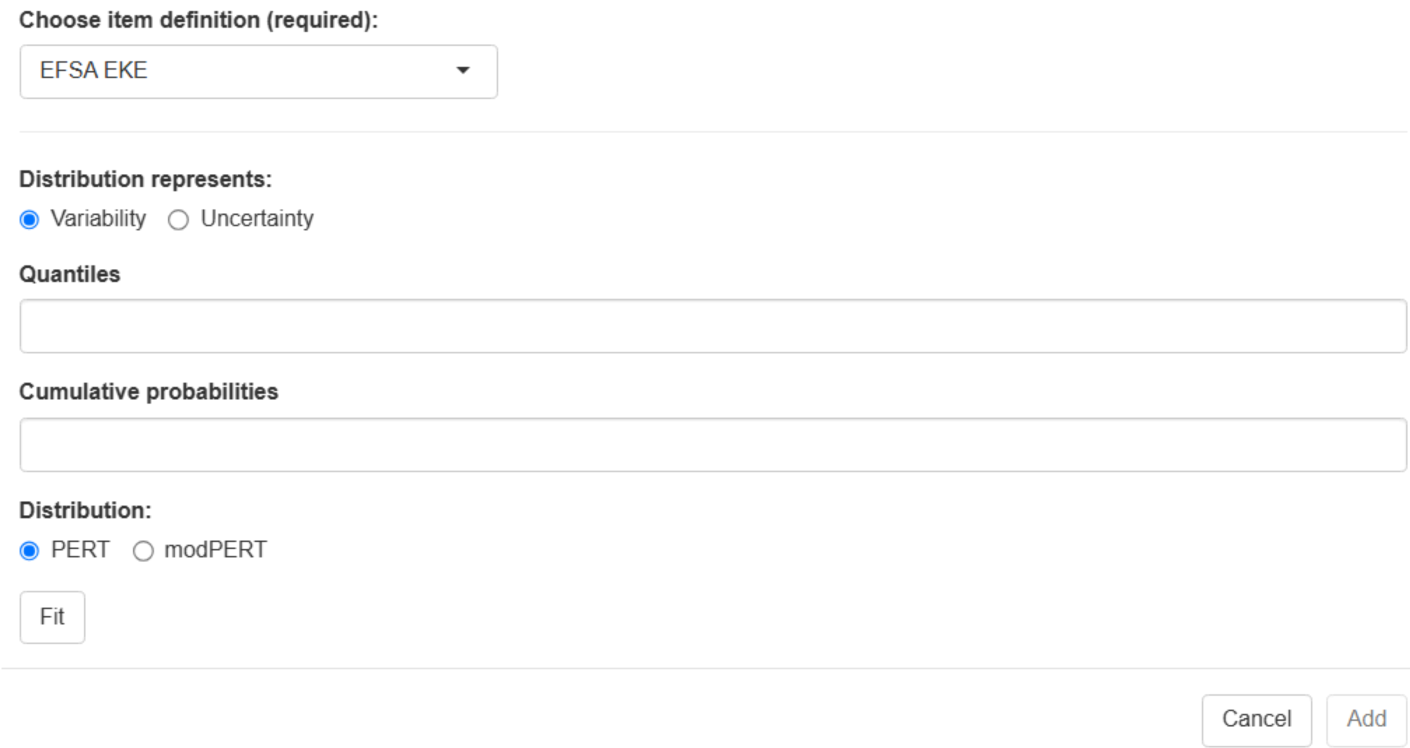


Figure 17: EFSA EKE node for defining a Pert distribution node.

Choose node definition	to be updated Select EFSA EKE.
Distribution represents	Select either variability or uncertainty.
Quantiles	Enter any number of quantiles, comma-separated.
Cumulative probability	Enter one cumulative probability for each of the quantiles, beginning with the value zero, comma-separated.

```
d -> choose(3,3)
print(d)
```

Defining an EFSA EKE node in shiny rrisk

The European Food Safety Authority (EFSA) have described a process for eliciting expert knowledge (REF). The use of EFSA EKE node involves three steps.

1. Conduct of the EFSA EKE protocol, which includes the definition of a target quantity, elicitation protocol and results in the descdocumenting the data source and defining node names and corresponding statistics.
2. A distribution node parameterised using input from an EFSA EKE may refer to variability or uncertainty, depending on the question formulation of the EKE process. The final result of the EKE consists of five percentiles that to be entered manually for fitting a Pert distribution (Figure 17). Details are illustrated using the example below.

Example for EFSA EKE (L. lignosellus in asparagus)

Read more

Node Table

After nodes are created, the following table will be created in shiny rrisk the information of each node in the model.

Search:

ID	Name	Group	Definition	Unit	MC Dim	Description	Actions
1	r	dose-response	pert(min = 1e-04, mode = 0.0012, max = 0.01)	1/CFU	U	frequency of infection by ingestion of one CFU	DELETE
2	R_rare	cooking	unif(min = 0.126, max = 1)	--	V	Reduction factor for cooking preference 'rare'. After cooking 100 % to 12.6 % of the bacteria survive.	DELETE
3	R_medium	cooking	unif(min = 0.04, max = 0.631)	--	V	Reduction factor for cooking preference 'medium'. After cooking 63.1 % to 4 % of the bacteria survive.	DELETE
4	R_well_done	cooking	unif(min = 0.002, max = 0.063)	--	V	Reduction factor for cooking preference 'well-done'. After cooking 6.3 % to 0.2 % of the bacteria survive.	DELETE

Node table created by shiny rrisk after the creation of nodes

The table contains the node ID, a sequence number according to the order a node was created during the the model creation. The rest of the columns present the information of each node that was entered during the node creation.

The table allows the user to sort each column separately, delete a node and search for particular terms. This later is very helpful when working with large models.

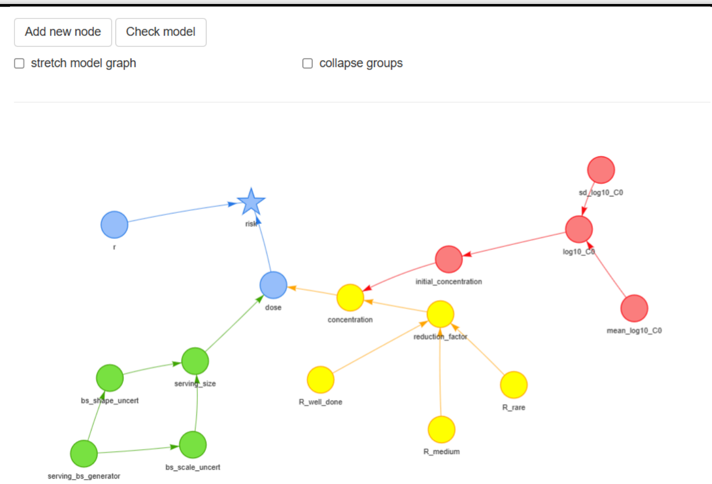
Graph

As soon as a node is created, a shape (square, circle or star) with a given color will appear in the screen. The star is the node representing the outcome(s), or the node that has no dependent nodes (or "child" nodes).

@MG/RO [- define Circle represents and squares - Is there any standards?]

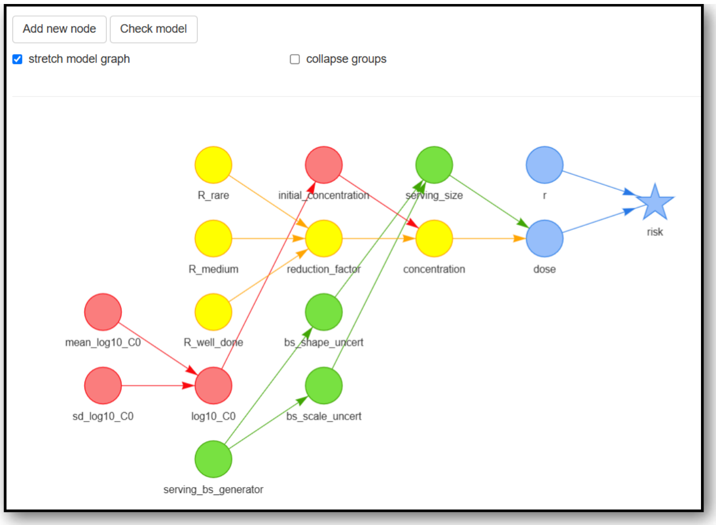
There are two check boxes the user can choose to change the layout of the graph created by shiny rrisk.

The user can leave unchecked the option "stretch model graph" so all the shapes can be moved freely by clicking with the mouse on a node. Otherwise by selecting "stretch model graph" all the shapes will be automatically organized with the"outcome" node located at the far right. If this option is selected, the user can only move the shapes up and down. This feature will help organizing very quickly the model's graph and improve the visualization, especially for models with large number of nodes.

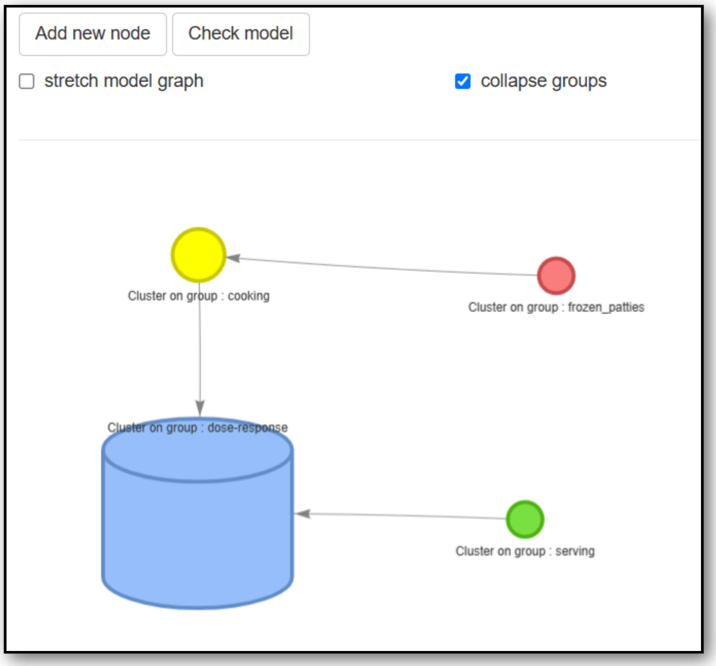


E. coli in hamburger meat model. Graph with stretch model graph unchecked

The next figure depicts the same model but with stretch model graph checked



The other check box is "collapse groups". If checked, all the nodes identified with the same "group name" will be hidden under the node identified with that group name. This is very useful when presenting a model with many nodes.



The user can click the mouse in any place on the screen that occupies the graph in order to move the whole graph, also the user can zoom in/zoom out using the appropriated combination of keys according to operating system (eg. Windows or MAC) or using the mouse. The user can double click on a node to open the node definition window (explained above).

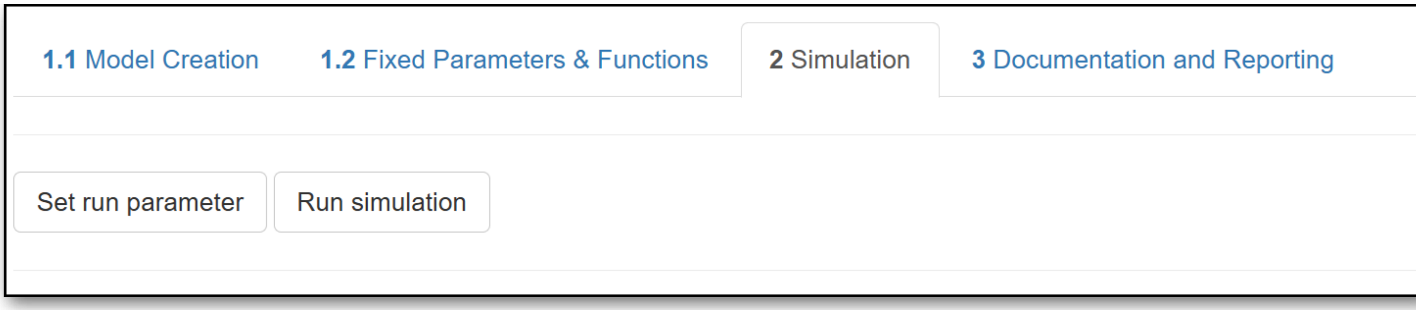
In order to move down to the definition table, the user should move the mouse out of the graph window or use the siding bar on the screen.

Check Model

Once all the nodes have been properly defined, the user should select Check Model for model verification. The model verification features assures that all the nodes are linked and properly defined. An error will appear other wise. The user will need to check that there is no isolated nodes (which could be due to typo in the node's names and/or an invalid equation)

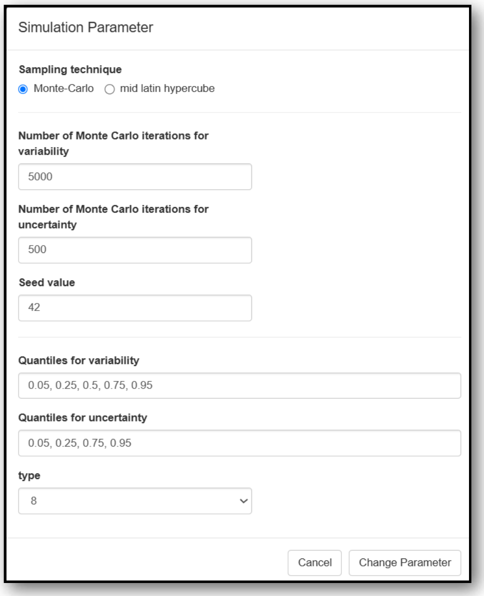
Simulation

> **Note**
>
> **This section will introduce you to:**
>
> - Obtain model results after creating and checking a model
> - Familiarize with the shiny rrisk model result interface
> - Interpret model results
> - Interpret sensitivity analysis



@Robert - change set run parameter to Set simulation parameters

Before executing (run) the simulation, user can set up several parameters by clicking **Set run parameter**. After setting these parameters, click **Change Parameter.**



SAMPLING TECHNIQUE	Select the random sampling method
NUMBER OF MONTE CARLO SIMULATIONS: VARIABILITY and UNCERTAINTY	Enter here the number of Monte Carlo random samples required for the simulation
QUANTILES: VARIABILITY and UNCERTAINTY	Enter here the percentiles for summarizing model results in the variability (1D) and uncertainty (2D)
SEED VALUE	Enter the integer number, this will allow to reproduce the same results
TYPE	?@Robert

After setting all the parameters, select **Run Simulation.** After the simulation is completed, shiny rrisk will present the another screen, with the **general results** (by default) and **sensitivity plot.**

@Robert - change to Sensitivity Analysis [change names in boxes to upper case to be consistent with other boxes]

Memory
The shiny rrisk server allocates a maximum memory of @@@ ro run the simulation. The amount of memory used is influenced by the number of nodes and the number of number samples required in the variability and uncertainty dimensions.

Good practice - Number of MC samples

Read more - Sampling methods

General Results

The results of all the nodes can be visualized from the drop down menu under Node. By default, shiny rrisk will start with the final node (eg. star shape node - or outcome).

Numerical result table

The output table will provide the summary statistics of each node according to the dimensions in variability and uncertainty. In addition, shiny rrisk will provide three plots in the natural scale or log10 transformed scale for summarizing the results.

One dimension Monte Carlo Model (1D MC) - Only Variability

The table below shows the results of the one dimension simulation for the E. coli in beef patties model after selecting 5000 random samples.

general results | sensitivity plot

Node

risk

mean	median	sd	5%	25%	50%	75%	95%	unit
0.191	0.0183	0.322	0	0.00088	0.0183	0.2	1	none

For 1D models, the results will be presented in only one row representing the variability of the node.

Two dimensions Monte Carlo Models (2D MC)- Variability and Uncertainty

The table below shows the results of the two dimension simulation for the E. coli in beef patties model after selecting 5000 and 500 random samples for the variability and uncertainty dimensions, respectively.

general results | sensitivity plot

Node

risk

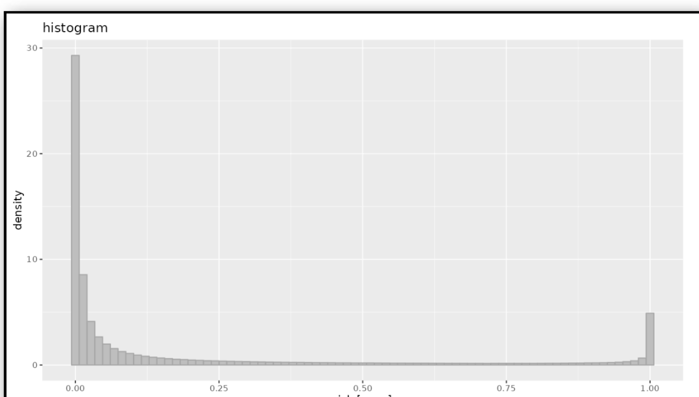
	mean	median	sd	5%	25%	50%	75%	95%	unit
mean	0.19	0.0303	0.305	0	0.00251	0.0303	0.256	0.91	none
median	0.189	0.0204	0.316	0	0.00135	0.0204	0.2	0.999	none
sd	0.0777	0.0366	0.0647	0	0.00403	0.0366	0.207	0.183	none
5%	0.0666	0.00256	0.18	0	0	0.00256	0.027	0.442	none
25%	0.135	0.00956	0.265	0	0	0.00956	0.0935	0.923	none
75%	0.243	0.0361	0.355	0	0.0033	0.0361	0.367	1	none
95%	0.314	0.0873	0.393	0	0.00907	0.0873	0.703	1	none

For 2D models, the summary statistics are presented in a matrix format, where rows represent uncertainty and the columns, the variability of the node.

Summary Plots

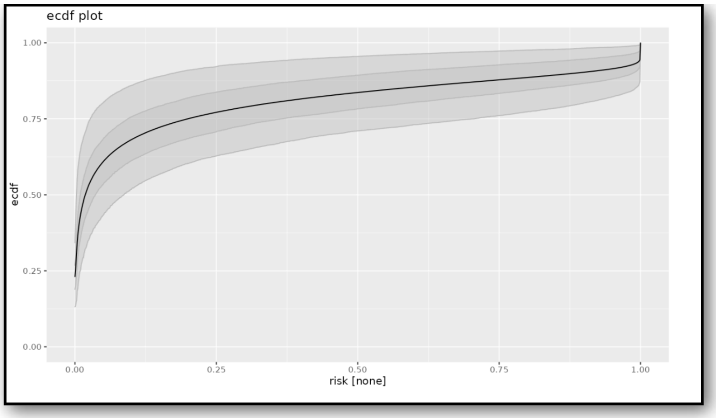
Histogram

This plot depicts the density distribution of the random values for a given node.



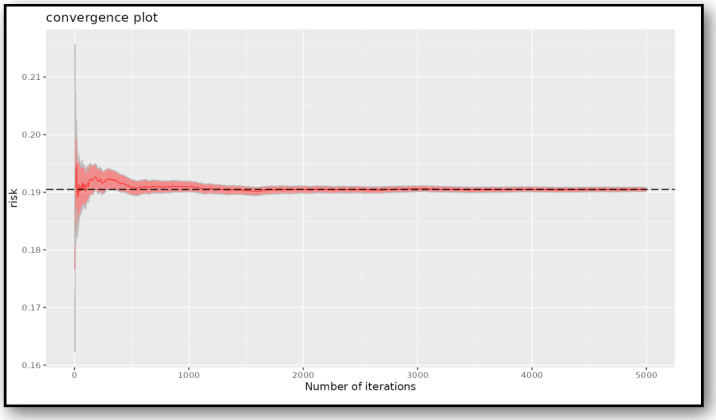
ECDF plot

This is the empirical cumulative density function plot, it is presented as an ascending cumulative plot. See read more for the proper interpretation of the graph.



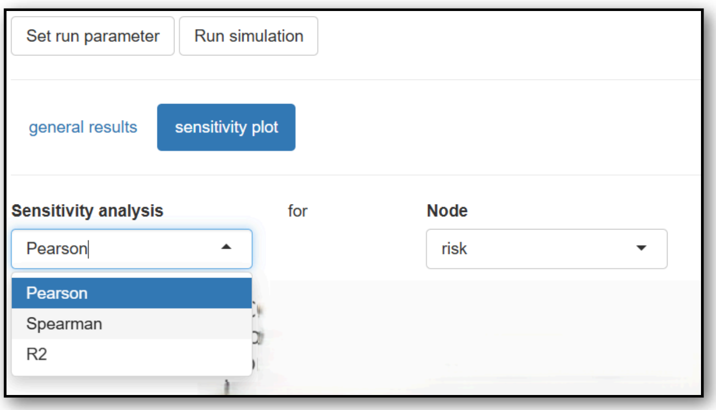
Convergence plot

@Robert - how to interpret this one?



Read more - 2d MC Models

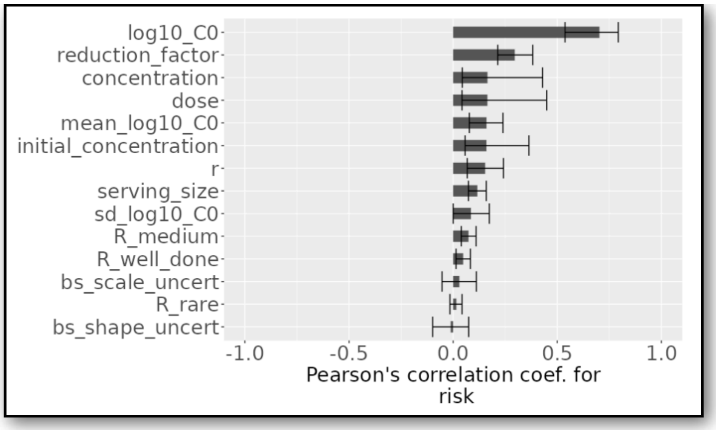
Sensitivity Plot



The sensitivity analyses results can be visualized by selecting the **sensitivity plot** tab in the main screen after running the simulation. Shiny rrisk will produce a tornado plot using two stasticts, “Pearson correlation coefficient” and “Spearman rho statistic” (the R2 statistic will be implemented in future versions of shiny rrisk).

The following plots presents the 2D tornado plot using the Pearson statistic for the E. coli model. All the input variables are postively correlation withe outcome (eg. the larger is the value of the input, the larger is the risk). However there are major differences in the influence of each input variable on the outcome. For instance the initial concentration of the pathogen (log10_C0) has a Pearson correlation coefficient of approximately 0.6 followed by the reduction facors and concentration.

Since this is a 2D model, the tornado plot also present the 95%. probability interval due to the uncertainty of the risk outcome.



Read more: Tornado plots

Save

Under this menu you can save the **Model** (*.rrisk) to your working device as explained before.

The **Report** option will generate a word document report with all the information entered in shiny rrisk.

The **Results** option will generate datasets in “*.csv” format [@Robert **- no working so not sure what I am getting**]

Documentation and Reporting

This tab is left to add additional information about the model authors and a general description of the model (eg. question of interest and model scope, methodology used, resources, etc).

Good practice: Developing a report

Greiner M, Sanchez J, and who else contributes, 2025